

Rhombix Technologies

DevOps Internship — Task Submission

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Task Selected	Project 1: Linux Basics & Command-Line Practice
Organization	Rhombix Technologies

1. Task Overview

This report documents the completion of Project 1 — Linux Basics & Command-Line Practice — assigned as part of the Rhombix Technologies DevOps internship. The objective of this task was to build a strong foundation in Linux, which forms the backbone of DevOps operations.

The task required practicing essential terminal commands, including file and directory navigation, file creation and editing, permission management, system resource monitoring, and package management, using an Ubuntu Linux environment.

Key Learning Objectives

- Navigating the Linux file system using commands such as `ls`, `cd`, and `pwd`.
- Creating and managing directories and files using `mkdir`, `touch`, `mv`, and `cp`.
- Viewing and sorting directory contents with `ls` flags (`-a`, `-l`, `-t`, `-r`).
- Installing and updating software packages using `apt-get` and `apt`.
- Monitoring system resources and running processes using `top` and `ps`.
- Checking system information using `uname`, `hostname`, `whoami`, and `df -h`.
- Reviewing command history using the `history` command.

2. Task Execution — Screenshots

Below are the screenshots captured while performing the assigned Linux practice task on an Ubuntu virtual machine.

2.1 Updating System Packages

The `apt upgrade` command was run to update installed packages on the system, demonstrating package management in Linux.

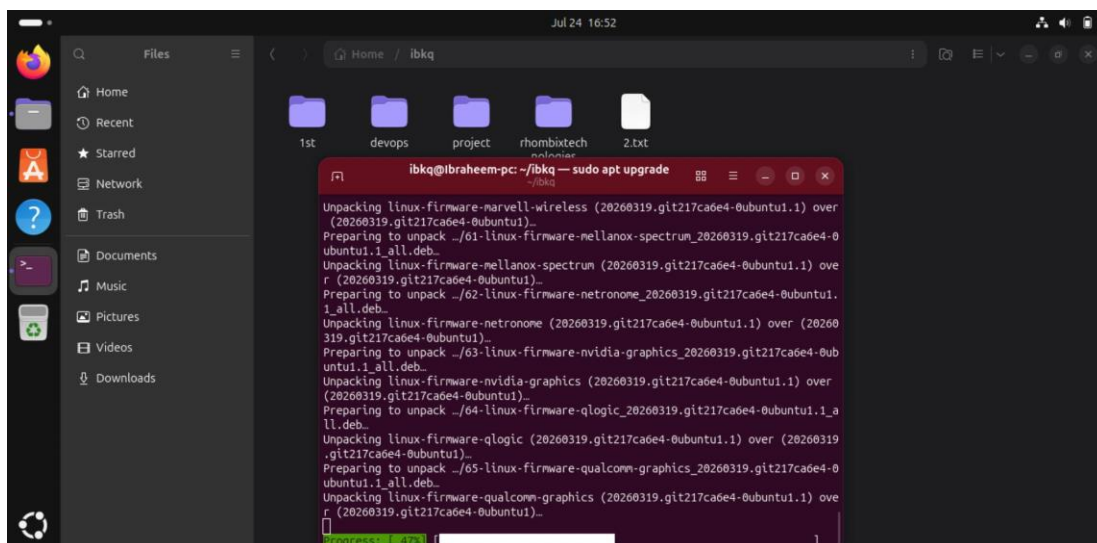


Fig 1: Running `sudo apt upgrade` — package unpacking in progress

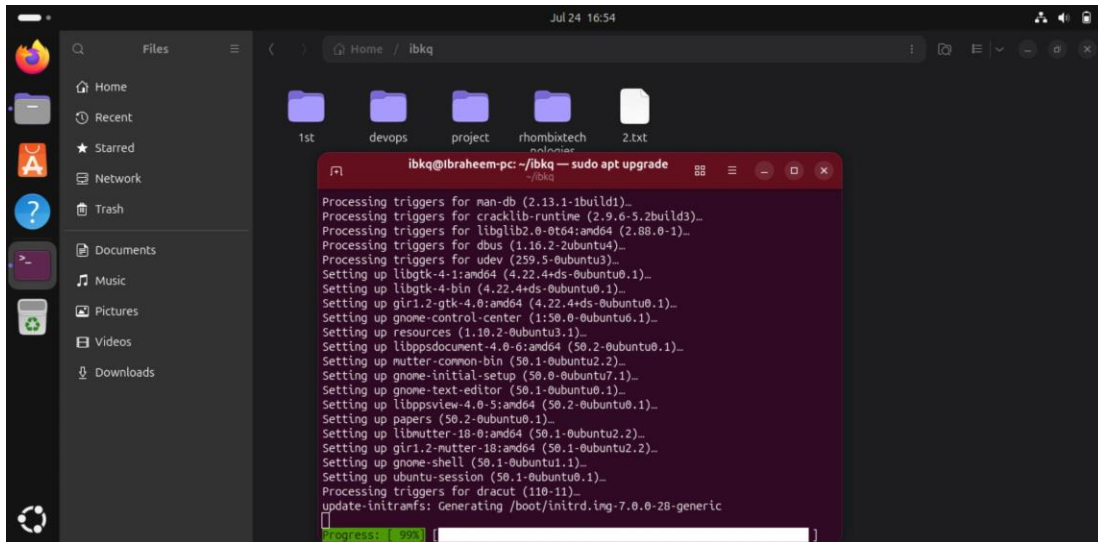


Fig 2: apt upgrade completing successfully at 99% progress

2.2 Directory Structure and Command History

Project directories (1st, devops, project, rhombixtechnologies) were created inside the home folder, and files were organized using mkdir, touch, mv, and cp. The history command was used to review all previously executed commands.

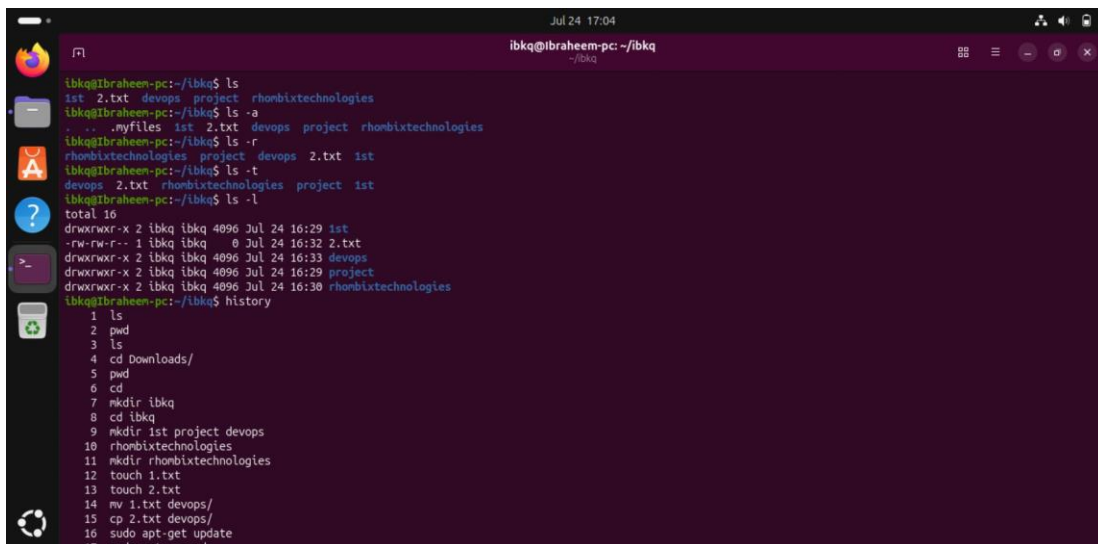
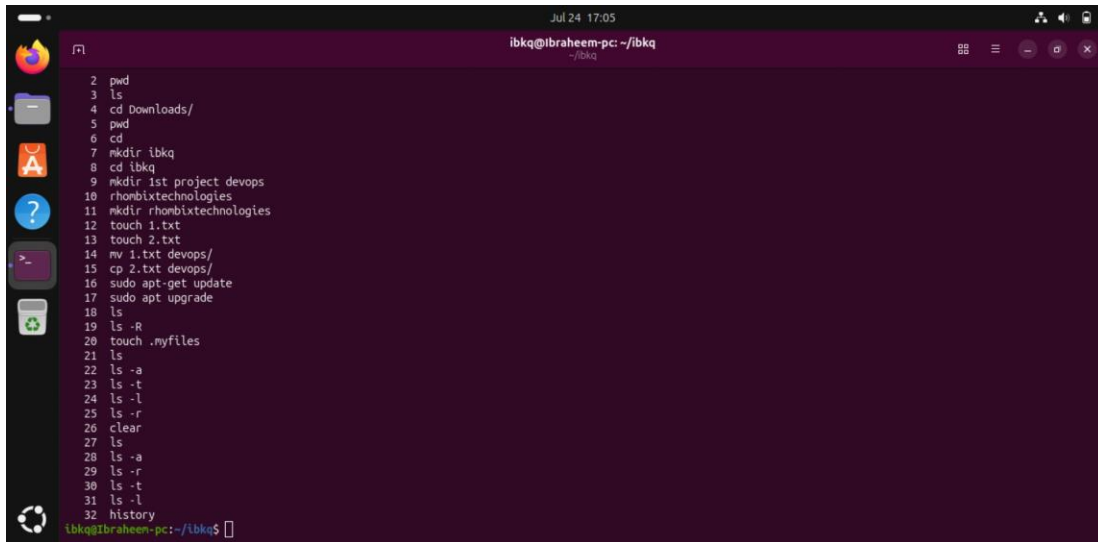


Fig 3: Directory listing (ls -a, -r, -t, -l) and command history

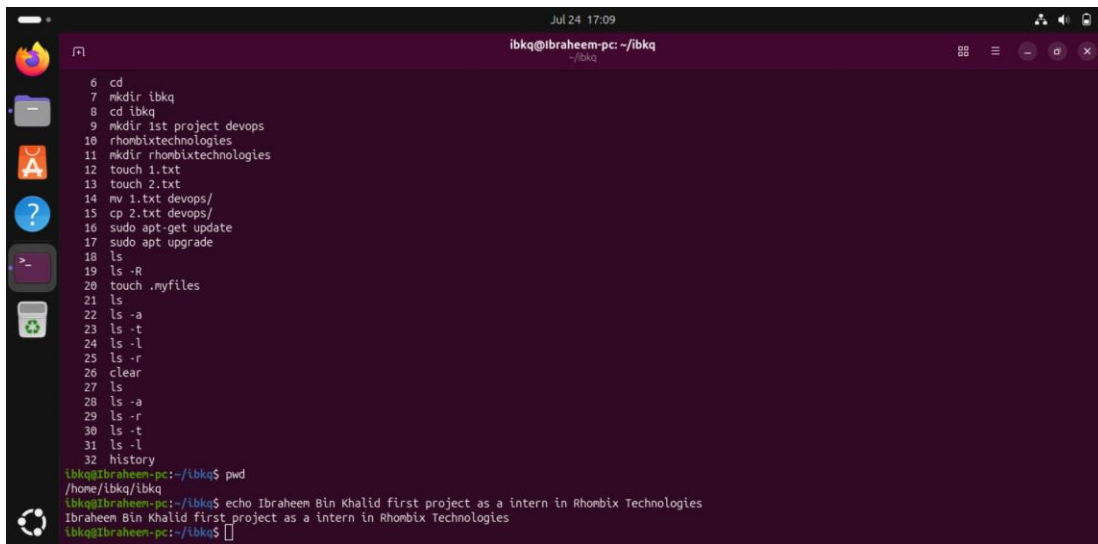
A terminal window titled 'ibkq@ibraheem-pc: ~/ibkq' with a date and time of 'Jul 24 17:05'. The terminal displays a list of 32 commands in a light blue font on a dark background. The commands include directory navigation, file creation, and system updates. The prompt 'ibkq@ibraheem-pc:~/ibkq\$' is visible at the bottom.

```
2 pwd
3 ls
4 cd Downloads/
5 pwd
6 cd
7 mkdir ibkq
8 cd ibkq
9 mkdir 1st project devops
10 rhombixtechnologies
11 mkdir rhombixtechnologies
12 touch 1.txt
13 touch 2.txt
14 mv 1.txt devops/
15 cp 2.txt devops/
16 sudo apt-get update
17 sudo apt upgrade
18 ls
19 ls -R
20 touch .myfiles
21 ls
22 ls -a
23 ls -t
24 ls -l
25 ls -r
26 clear
27 ls
28 ls -a
29 ls -r
30 ls -t
31 ls -l
32 history
ibkq@ibraheem-pc:~/ibkq$
```

Fig 4: Full command history showing directory and file operations

2.3 File Content Verification

The pwd command confirmed the current working directory, and the echo command was used to write a short introductory message identifying the intern and the project.

A terminal window titled 'ibkq@ibraheem-pc: ~/ibkq' with a date and time of 'Jul 24 17:09'. It shows the same command history as Figure 4, followed by the execution of 'pwd' and 'echo'. The 'pwd' command outputs '/home/ibkq/ibkq', and the 'echo' command outputs a message identifying the user and project. The prompt 'ibkq@ibraheem-pc:~/ibkq\$' is visible at the bottom.

```
6 cd
7 mkdir ibkq
8 cd ibkq
9 mkdir 1st project devops
10 rhombixtechnologies
11 mkdir rhombixtechnologies
12 touch 1.txt
13 touch 2.txt
14 mv 1.txt devops/
15 cp 2.txt devops/
16 sudo apt-get update
17 sudo apt upgrade
18 ls
19 ls -R
20 touch .myfiles
21 ls
22 ls -a
23 ls -t
24 ls -l
25 ls -r
26 clear
27 ls
28 ls -a
29 ls -r
30 ls -t
31 ls -l
32 history
ibkq@ibraheem-pc:~/ibkq$ pwd
/home/ibkq/ibkq
ibkq@ibraheem-pc:~/ibkq$ echo Ibraheem Bin Khalid first project as a intern in Rhombix Technologies
Ibraheem Bin Khalid first project as a intern in Rhombix Technologies
ibkq@ibraheem-pc:~/ibkq$
```

Fig 5: Verifying working directory and printing intern details with echo

2.4 System Resource Monitoring

The top command was used to monitor real-time CPU and memory usage, running processes, and system load average.

```
Jul 24 17:16
ibkq@ibraheem-pc: ~/ibkq — top
top - 17:16:06 up 1:12, 1 user, load average: 0.23, 0.11, 0.18
Tasks: 214 total, 1 running, 212 sleeping, 0 stopped, 1 zombie
%Cpu(s): 10.7 us, 3.2 sy, 0.0 ni, 85.9 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 3398.8 total, 523.8 free, 1612.2 used, 1569.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 1786.6 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
  9784 ibkq       20   0 4049272 444572 165076 S   20.7  12.0   1:55:57 gnome-shell
 11781 ibkq       20   0 1809840 400792 143000 S    2.6  11.5   0:54:42 ptxixs
      1 root        0   0 26108 17292 11724 S    1.3   0.5   0:09:34 systemd
 1402 message+  20   0 12260 9152  4872 S    1.3   0.3   0:04:77 dbus-daemon
 1423 polkitd  20   0 324620 12644  8460 S    1.0   0.4   0:02:12 polkitd
 25464 ibkq       20   0 22064 6424  4244 R    1.0   0.2   0:00:13 top
 1440 root        0   0 18516 9744  8288 S    0.3   0.3   0:01:06 systemd-logind
 9063 ibkq       20   0 470592 11832  7808 S    0.3   0.3   0:02:01 ibus-daemon
10844 ibkq       20   0 160264 2400  2036 S    0.3   0.1   0:00:00 VBoxClient
 11038 root        0   0 0 0 0 I    0.3   0.0   0:01:90 kworker/u9:0-events_unbound
 25284 root        0   0 0 0 0 I    0.3   0.0   0:00:49 kworker/0:0-mm_percpu_wq
 25312 root        0   0 0 0 0 I    0.3   0.0   0:00:78 kworker/u9:3-flush-8:0
      2 root        0   0 0 0 0 S    0.5   0.0   0:00:01 kthreadd
      3 root        0   0 0 0 0 S    0.5   0.0   0:00:00 pool_workqueue_release
      4 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/R-rcu_gp
      5 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/R-sync_wq
      6 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/R-kvfree_rcu_reclaim
      7 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/R-slub_flushwq
      8 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/R-nets
     12 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/u8:0-ipv6_addrconf
     13 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker/R-mm_percpu_wq
     14 root        0 0 0 0 0 S    0.0   0.0   0:00:12 ksoftirqd/0
     15 root        0 0 0 0 0 I    0.0   0.0   0:02:12 rcu_preempt
     16 root        0 0 0 0 0 S    0.0   0.0   0:00:00 rcu_exp_par_gp_kthread_worker/0
     17 root        0 0 0 0 0 S    0.0   0.0   0:00:02 rcu_exp_gp_kthread_worker
```

Fig 6: top command showing live system resource usage

2.5 Process Management

The ps and ps -a commands were used to list currently running processes and their process IDs (PIDs).

```
Jul 24 17:21
ibkq@ibraheem-pc: ~/ibkq
Tasks: 214 total, 1 running, 212 sleeping, 0 stopped, 1 zombie
%Cpu(s): 1.0 us, 0.3 sy, 0.0 ni, 98.5 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3398.8 total, 515.7 free, 1620.0 used, 1562.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 1778.8 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 11781 ibkq       20   0 1801816 393912 136120 S   1.7  11.3   0:59:07 ptxixs
 25464 ibkq       20   0 22064 6424  4244 R    0.7   0.2   0:00:30 top
     62 systemd+  20   0 16000 7760  6716 S    0.3   0.2   0:02:23 systemd+
    8156 root        0   0 350152 2540  2264 S    0.3   0.1   0:04:71 VBoxDRM+
  9784 ibkq       20   0 4042008 444464 177904 S    0.3  12.8   2:05:44 gnome-s+
 11804 ibkq       20   0 240044 8024  7308 S    0.3   0.2   0:01:18 ptxixs+
      1 root        0   0 26108 17292 11724 S    0.5   0.5   0:09:35 systemd
      2 root        0   0 0 0 0 S    0.5   0.0   0:00:01 kthreadd
      3 root        0   0 0 0 0 S    0.5   0.0   0:00:00 pool_wor
      4 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker+
      5 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker+
      6 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker+
      7 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker+
      8 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker+
     12 root        0 0 0 0 0 I    0.0   0.0   0:00:00 kworker+
     13 root        0 -20 0 0 0 I    0.0   0.0   0:00:00 kworker+
     14 root        0 0 0 0 0 S    0.5   0.0   0:00:12 ksoftir+

ibkq@ibraheem-pc:~/ibkq$ ps
  PID TTY          TIME CMD
 11836 pts/0        00:00:00 bash
 25484 pts/0        00:00:00 ps

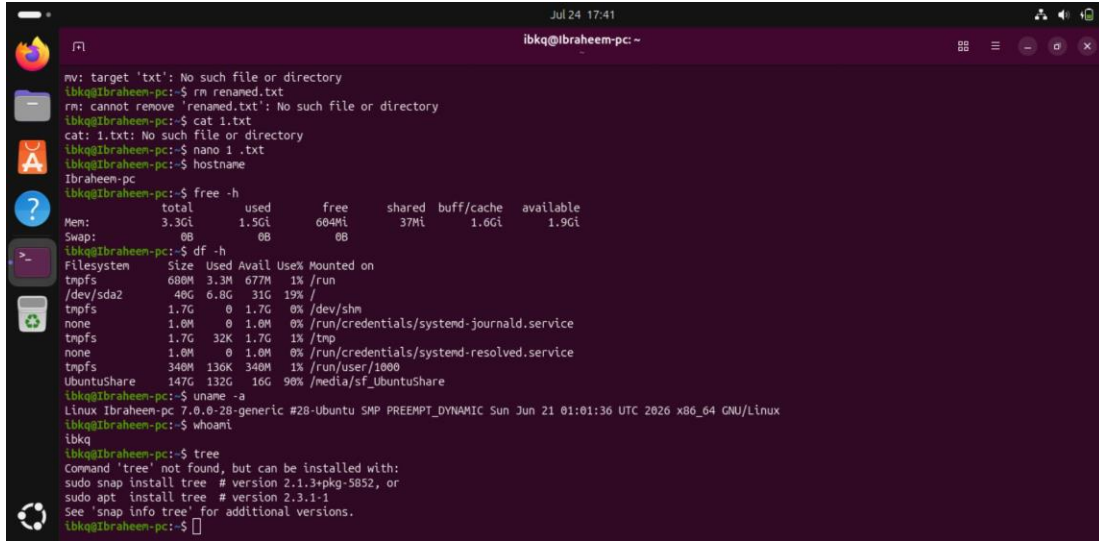
ibkq@ibraheem-pc:~/ibkq$ ps -a
  PID TTY          TIME CMD
   940 tty2        00:00:00 gnome-session-i
 25500 pts/0        00:00:00 ps

ibkq@ibraheem-pc:~/ibkq$
```

Fig 7: Listing running processes using ps and ps -a

2.6 System Information Commands

Additional system information commands were practiced, including free -h (memory usage), df -h (disk usage), uname -a (kernel/system info), whoami (current user), and hostname.

A terminal window titled 'ibkq@ibraheem-pc: ~' with a date and time of 'Jul 24 17:41'. The terminal shows a series of commands and their outputs. The commands include: 'mv: target \'.txt\': No such file or directory', 'ibkq@ibraheem-pc:~\$ rm renamed.txt', 'rm: cannot remove \'.txt\': No such file or directory', 'ibkq@ibraheem-pc:~\$ cat 1.txt', 'cat: 1.txt: No such file or directory', 'ibkq@ibraheem-pc:~\$ nano 1.txt', 'ibkq@ibraheem-pc:~\$ hostname', 'ibraheem-pc', 'ibkq@ibraheem-pc:~\$ free -h', 'ibkq@ibraheem-pc:~\$ df -h', 'ibkq@ibraheem-pc:~\$ uname -a', 'ibkq@ibraheem-pc:~\$ whoami', 'ibkq', and 'ibkq@ibraheem-pc:~\$ tree'. The outputs include memory usage statistics, disk usage statistics, system information, and user information.

```
mv: target '.txt': No such file or directory
ibkq@ibraheem-pc:~$ rm renamed.txt
rm: cannot remove '.txt': No such file or directory
ibkq@ibraheem-pc:~$ cat 1.txt
cat: 1.txt: No such file or directory
ibkq@ibraheem-pc:~$ nano 1.txt
ibkq@ibraheem-pc:~$ hostname
ibraheem-pc
ibkq@ibraheem-pc:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.3Gi        1.5Gi        604Mi        37Mi        1.6Gi        1.9Gi
Swap:              0B              0B
ibkq@ibraheem-pc:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            680M  3.3M  677M   1% /run
/dev/sda2        40G   6.8G   31G  19% /
tmpfs            1.7G   0 1.7G   0% /dev/shm
tmpfs            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs            1.7G  32K  1.7G   1% /tmp
tmpfs            1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           340M 136K  340M   1% /run/user/1000
UbuntuShare     147G  132G   16G  90% /media/sf_UbuntuShare
ibkq@ibraheem-pc:~$ uname -a
Linux Ibraheem-pc 7.0.0-28-generic #28-Ubuntu SMP PREEMPT_DYNAMIC Sun Jun 21 01:01:36 UTC 2026 x86_64 GNU/Linux
ibkq@ibraheem-pc:~$ whoami
ibkq
ibkq@ibraheem-pc:~$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5052, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
ibkq@ibraheem-pc:~$
```

Fig 8: System information — memory, disk usage, kernel details, and user info

3. Conclusion

Through this task, hands-on experience was gained in essential Linux command-line operations that are fundamental to DevOps workflows. Practicing file management, package installation, process monitoring, and system diagnostics has provided a solid foundation for working with Linux servers in real-world DevOps environments.

This task has been completed as part of the Rhombix Technologies DevOps Internship Program and is being submitted in accordance with the internship submission guidelines.